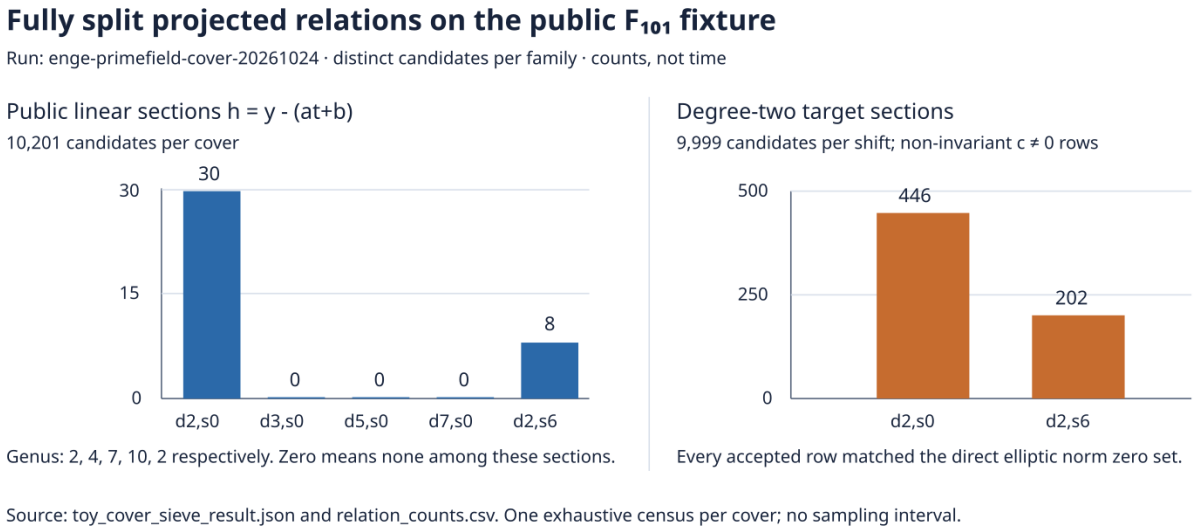
Native Replay Record: Prime-Field Cover Sections

Run enge-primefield-cover-20261024-native-2315Z

24 October 2026, 23:15 UTC

Host and isolation. Intel Xeon 6975P-C, x86-64, 32 logical CPUs, shared EC2 host. No CPU pinning or isolated timing claim. One exhaustive census was requested and completed for each of $(d, s) = (2, 0), (3, 0), (5, 0), (7, 0), (2, 6)$. The input is the public $E/\mathbb{F}_{101} : y^2 = x^3+x+46$ with $G = (2, 37)$ and $Q = (64, 19)$. The code uses deterministic loops and rho salts. The command and raw JSON record are named in the adjacent `REPORT.md`.



Observed counts. The five public-section split counts were $30, 0, 0, 0, 8$ of 10,201 candidates per cover. The two degree-two target-section native counts were 446 and 202 of 9,999 candidates per shift. Every one of the 202 accepted $(2, 6)$ rows matched the direct elliptic norm zero set. Target-only streaming found the verified scalar 37 after 1,070 candidates, 108,070 cover-point tests, and 22 accepted rows. The signed matrix rank was 21 on 22 columns. The deterministic rho control required 72 elliptic group operations through collision.

Scope. These are complete deterministic counts for the stated families and one run per cover, with no sampling interval. The units of the cover and rho counts differ; wall time, memory, setup and shift selection were not measured. No speed ratio is supported. The root `REPORT.pdf` contains the algebra and interpretation.